

Lecture 3

SQA system

Outline

- SQA system
- SQA components
- Considerations for SQA system construction

References

- Daniel Galin, Software Quality Assurance: From theory to implementation, chapter 4.1–4.8

SQA system

What is SQA system?

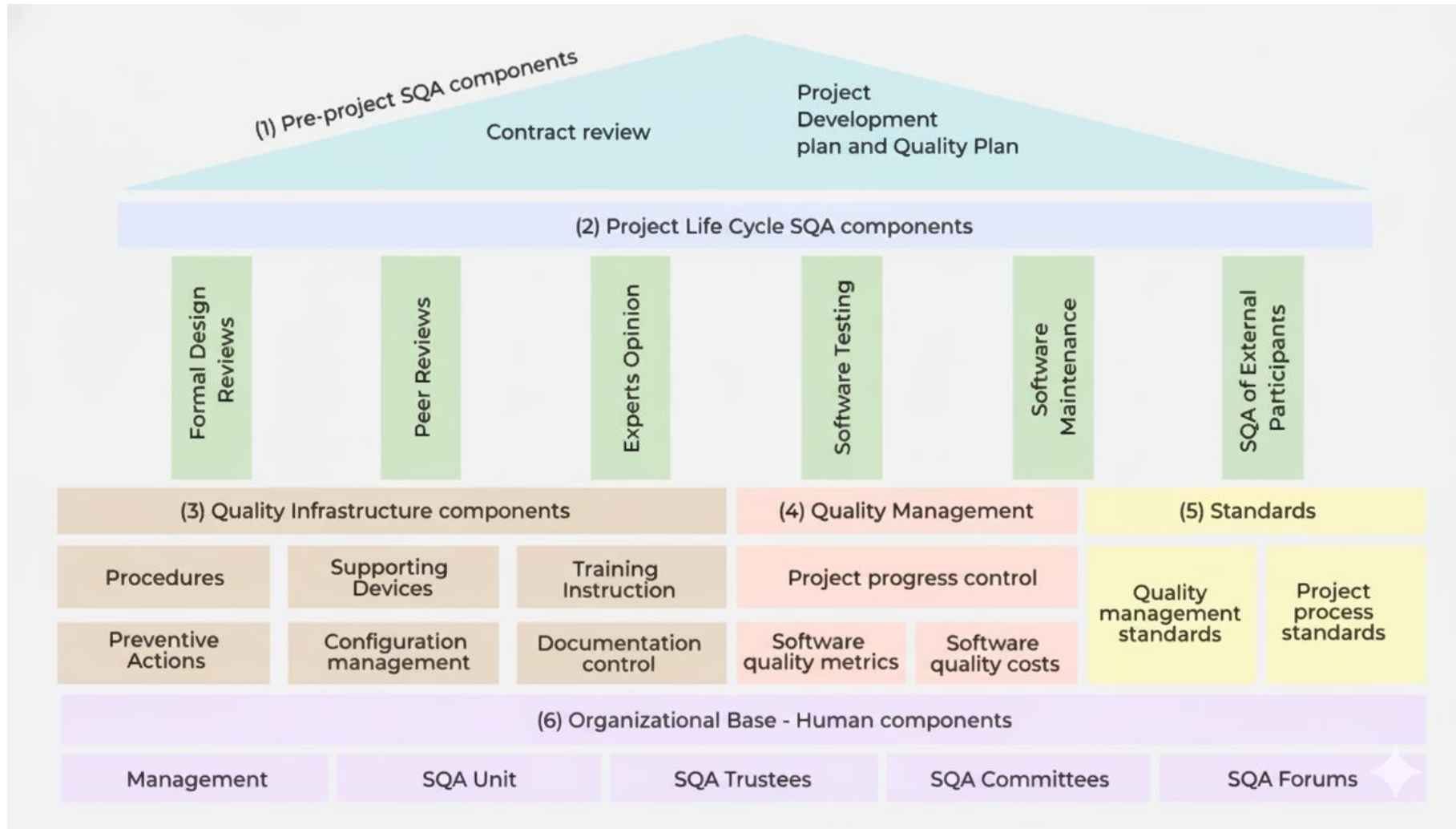
*“An integrated framework that combines a wide range of **SQA components**, designed to challenge the multitude of sources of software errors and to achieve an acceptable level of software quality”*

SQA system components

SQA system components are divided into 6 classes:

1. Pre-project components
2. Components of project life cycle activities assessment
3. Components of infrastructure error prevention and improvement
4. Components of software quality management
5. Components of standardization, certification, and SQA system assessment
6. Organizing for SQA – the human components

SQA System - SQA architecture



Pre-project components

To to improve the preparatory steps taken prior to initiating work on the project itself

Divided into two SQA components:

- **Contact reviews:** includes a detailed examination of the project proposal draft and the contract drafts
- **Development and quality plans:** Once the contract is signed, a plan is prepared of the project and its integrated quality activities

Pre-project components

- Development Plan: the main issues treated in the project development plan are:
 - Schedules
 - Required manpower and hardware resources
 - Risk evaluations
 - Organizational issues: team members, subcontractors and partnerships
 - Project methodology, development tools,...
 - Software reuse plans

Pre-project components

- Quality Plan: the main issues treated in the project's quality plan are:
 - Quality goals, expressed in the appropriate measurable terms
 - Criteria for starting and ending each project stage
 - List of reviews, tests, and other scheduled verification and validation activities

Software life cycle components

Several SQA components enter the software development project life cycle at different points.

Software project life cycle components:

- Reviews
- Expert opinions
- Software testing
- Software maintenance components
- Assurance of the quality of the subcontractors' work and the customer supplied parts.

Project life cycle components

- **Review:** Focus on documents and code that produced in development process.
- Reviews can be categorized as follows:
 - **Formal design reviews (DRs):** These documents require formal professional approval of their quality before moving to the next step. Committees are composed of senior professionals including the project leader
 - **Peer reviews (Inspections and Walkthroughs):** Directed at reviewing short documents. Reviewers are all peers. The main objective is to detect as many design and programming faults as possible

Project life cycle components

- **Expert opinions:** Introduce additional external capabilities into the organization:
 - Outside experts may join a DR committee
 - Their opinions may replace a DR(design review)
 - In case of disagreement of senior professionals, an outside expert may support a decision
- **Software testing:** Includes formal SQA components that are targeted toward review of the actual running of the software. All tests must be designed, planned and approved.

Project life cycle components

Software maintenance components:

- Pre-maintenance components:
- Software development life cycle components
- Infrastructure SQA components
- Managerial control SQA components

Project life cycle components

Assurance of the quality of external participants' work:

- In case transfer a part of work to subcontractors
- Also check the quality of the hardware, software, staff and training supplied by the customer

Infrastructure components for error prevention and improvement

To eliminate or make the errors gone missing or well, at least reduce the rate of errors. It's applied to entire organization

This class of SQA components includes:

- Procedures and work instructions
- Templates and checklists
- Staff training, retraining, and certification
- Preventive and corrective actions
- Configuration management
- Documentation control

Infrastructure components for error prevention and improvement

- **Procedures and work instructions:** These components translate organizational knowledge into standard routines to ensure effective quality results
 - **Procedures:** Generally applicable definitions for development activities designed to serve the **entire organization**
 - **Work Instructions:** Highly detailed directions for specific methods, typically used by **specialized teams** in unique instances

Infrastructure components for error prevention and improvement

- **Templates and checklists:** These tools combine higher efficiency with higher quality by leveraging accumulated professional experience
 - **Time Saving:** Reduces the effort needed to define document structures or review lists
 - **Completeness:** Ensures that all necessary subjects are covered during documents preparation and reviews
 - **Standardization:** Improves communication between development teams and review committees through uniform agendas and formats

Infrastructure components for error prevention and improvement

- **Staff training, retraining, and certification:** a well-instructed staff is the key to quality
 - **Training & Retraining:** Focuses on new employees and those changing assignments
 - **Continuous Updates:** Keeps the staff informed on professional developments and in-house hands-on experience
 - **Certification:** Formalizes the process by acknowledging employees only after they demonstrate specific knowledge and ability

Infrastructure components for error prevention and improvement

- **Preventive and corrective actions:** involves the systematic study of failure and success data
 - **Prevention:** Implementing changes to stop similar failures from occurring in the future
 - **Correction:** Fixing similar faults found across other projects or teams
 - **Success Replication:** Identifying and implementing successful methodologies to repeat high-quality outcomes

Infrastructure components for error prevention and improvement

- **Configuration management:** Software involves constant modifications (versions and releases) over several years. This component manages the hazards of simultaneous changes
 - **Change Control:** Introduces procedures to approve and record all modifications, preventing unauthorized changes to approved versions
 - **Version Control:** Uses computerized tools to manage version specifications for different installation sites

Infrastructure components for error prevention and improvement

- **Documentation control:** refer to customer requirements, contracts, design reports, and project plans. Documentation control activities entail:
 - Definition of the types of controlled documents needed
 - Specification of the formats, document identification methods, ...
 - Definition of review and approval processes for each controlled document
 - Definition of the archive storage methods

Management SQA components

These components support managerial control over software development projects and maintenance services

Control components include:

- Project progress control
- Software quality metrics
- Software quality costs

Management SQA components

- **Project progress control:** The main objective is the timely discovery of undesirable situations to allow for effective corrective measures. Project progress control focuses on area:
 - Resource usage
 - Schedules
 - Risk management activities
 - The budget

Management SQA components

- **Software quality metrics:** Measurement of the various aspects of software quality
 - **Scope of Application:**
 - Functional quality
 - Productivity
 - Organizational aspects
 - **Specific Metrics List for:**
 - Quality of software development and maintenance activities
 - Development teams' productivity
 - Help desk and maintenance teams' productivity
 - Software faults density
 - Schedule deviations

Management SQA components

- **Software quality costs:** Total Quality Cost = **Costs of Control + Costs of Failure**
 - **Costs of Control:** Prevention, appraisal, and managerial control costs
 - **Costs of Failure:** Internal, external, and managerial failure costs
 - **Managerial Strategy:**
 - **Optimization:** Increasing resources for control activities leads to much larger savings in failure costs, reducing the total quality cost
 - **Funding:** Management is more likely to support SQA improvements that prove profitable

SQA Standards, certification, and assessment components

Aim:

- Utilization of international professional knowledge.
- Improvement of coordination with other organizations' quality systems
- Objective professional evaluation and measurement

They are classified into 2 sub-classes:

- Quality management standards: SEI CMM, ISO 9001
- Project process standard: IEEE 1012, ISO/IEC 12207

Organizing for SQA — the human components

The SQA organizational base includes managers, testing personnel, the SQA unit, and practitioners (trustees, committee, and forum members)

Main objectives:

- Develop and support implementation of SQA components
- Detect deviations from SQA procedures and methodology
- Suggest improvements to SQA components

Considerations for SQA system construction

Considerations for SQA system construction

System Flexibility SQA systems are not "one-size-fits-all"

Two Main Decision Categories:

- The SQA organizational base
- The SQA components to be implemented within the organization and the extent of their use

Fundamental Influencing Factors:

- The organization
- Projects & Services
- Professional Staff

Summary

- SQA system
- SQA components
- Considerations for SQA system construction